

SUMAN MAITI

PhD Student, Department of Computer Science and Engineering, IIT Kharagpur

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EDUCATION

Indian Institute of Technology Kharagpur

Doctor of Philosophy (PhD) in Computer Science and Engineering

Kharagpur, India

Jan 2022 – Present

- **Research Focus:** Cyber-Physical Systems Security, Smart Grid Security, Formal Methods, Safe RL.
- **Coursework:** High Performance Computer Architecture, Foundations of Algorithm Design and Machine Learning, AI for Cyber-Physical Systems, Computational Foundations of CPS, Statistical Methods.

Kalyani Government Engineering College

Bachelor of Technology (B.Tech) in Electrical Engineering | DGPA: 9.03/10

Nadia, India

Sep 2017 – Aug 2021

D.A.V. Model School, IIT Campus

All India Senior School Certificate Examination (Class XII) | 93.8%

Kharagpur, India

Mar 2017

RESEARCH EXPERIENCE

Junior Research Fellow (JRF)

Indian Institute of Technology Kharagpur

Oct 2021 – Dec 2023

Kharagpur, India

- **Smart Grid Privacy and Security:** Conducted research into vulnerabilities in smart grid environments and their countermeasures.
- Applied machine learning techniques, particularly reinforcement learning, to identify vulnerabilities in IEEE 9, 14, 37, and 39 bus power grid models.
- Investigated industrial SCADA communication protocols including DNP3, DNP3-SA, and Modbus for threat modeling.

Summer Research Intern

Indian Institute of Technology Kharagpur

Jun 2019 – Jul 2019

Kharagpur, India

- **Design of a Digital Clock using 8085 Microprocessor:** Implemented microprocessor programming to display time using a seven-segment display, gaining foundational experience in low-level hardware-software interaction.

SELECTED PROJECTS

Formal Methods for Smart Grid Privacy and Security | *Staliro, RL, IEEE Bus Models*

- Investigated vulnerabilities of IEEE 9, 14, and 39 bus power grid generation and distribution units using a reinforcement learning agent combined with the formal falsification tool Staliro.
- Produced targeted attack sequences and formal safety proofs under Metric Temporal Logic (MTL) specifications.

Fault Studies of Wind-Farm Embedded Power Network | *MATLAB/Simulink*

- Simulated symmetrical and asymmetrical fault conditions in a 225/90 kV power transmission network.
- Analyzed voltage and power waveforms of transmission lines during fault conditions (Published at IEEE DevIC 2021).

CPU Architecture Performance Estimation | *GEM5 Simulator, C++*

- Developed a framework for generating different CPU configuration combinations from user-specified parameters.
- Evaluated performance by running benchmark programs on the GEM5 system simulator, enabling systematic architectural trade-off analysis.

Autonomous Driving in F-1/10 Vehicle | *Control Theory | In Progress*

- Designing a controller for an F-1/10 autonomous vehicle that enables it to follow a given track while avoiding obstacles with formal safety guarantees.

PUBLICATIONS

Journal Articles

- S. Ghosh, **S. Maiti**, D. Mukhopadhyay, S. Dey. "*Differentially Private Real Time Pricing Control for Smart Grids*", ACM Transactions on Cyber-Physical Systems, 2025.

Conference Proceedings

- **S. Maiti**, A. Balabhaskara, S. Adhikary, I. Koley, S. Dey. "*Targeted Attack Synthesis for Smart Grid Vulnerability Analysis*", Proceedings of the ACM SIGSAC Conference on Computer and Communications Security (CCS '23), 2023. [Cited by 18]
- **S. Maiti**, S. Som, S. Mondal, P. K. Gayen. "*Symmetrical and Asymmetrical Fault Studies of Wind-Farm Embedded Power Network*", Devices for Integrated Circuit (DevIC), IEEE, 2021.

Preprints

- **S. Maiti**, S. Adhikary, S. Dey, A. R. Hota. "*Learning-Enabled Adaptive Voltage Protection Against Load Alteration Attacks on Smart Grids*", arXiv:2411.15229, 2024.
- **S. Maiti**, S. Dey. "*Smart Grid Security: A Verified Deep Reinforcement Learning Framework to Counter Cyber-Physical Attacks*", arXiv:2409.15757, 2024. [Cited by 11]

TEACHING ASSISTANT EXPERIENCE (IIT KHARAGPUR)

Programming and Data Structures Laboratory Spring Semester 2026 Instructor: Prof. Dipanwita Roy Chowdhury	Jan 2026 – May 2026
Reinforcement Learning Autumn Semester 2025 Instructor: Prof. Soumyajit Dey	Jul 2025 – Jan 2026
Formal Language and Automata Theory Spring Semester 2025 Instructors: Prof. Soumyajit Dey & Prof. Animesh Mukherjee	Feb 2025 – Jul 2025
Programming and Data Structures Theory Autumn Semester 2024 Instructor: Prof. Soumyajit Dey	Jul 2024 – Jan 2025
Formal Language and Automata Theory Autumn 2023 – Spring 2024 Instructors: Prof. Soumyajit Dey & Prof. Animesh Mukherjee	Aug 2023 – Mar 2024
Programming and Data Structures Laboratory (3 Semesters) Instructors: Prof. Pallab Dasgupta, Prof. Soumyakanti Ghosh, Prof. K. S. Rao, Prof. Debasis Samanta	Oct 2022 – Jul 2023

AWARDS, GRANTS & TALKS

ACM IARCS Travel Grant: Awarded INR 1,00,000 in recognition of research contribution.

ACM CCS 2023 Student Travel Grant: Recipient of travel grant for the ACM Conference on Computer and Communications Security.

Invited Talk: Delivered talk at ACM ARCS Event 2023 at NISER Bhubaneswar on smart grid security and attack synthesis.

GATE 2021 (Electrical Engineering): Secured **94.4 Percentile** in the Graduate Aptitude Test in Engineering, qualifying for doctoral admissions.

TECHNICAL SKILLS

Programming Languages: Python, C++, C, PHP, LaTeX

ML / DL Frameworks: TensorFlow, Keras, Scikit-learn, Stable Baselines3, Gymnasium, PyTorch

Formal Verification: Z3 SMT Solver, Staliro (Falsification), UPPAAL, TuLiP, Metric Temporal Logic, Hybrid Automata

Engineering Tools: MATLAB, Simulink, RT-Lab, LabVIEW, Artemis, HYPERSIM, EMTP-RV / Ephasorsim

Hardware & Platforms: OP4510 (Opal-RT), SEL-421 (Schweitzer), Raspberry Pi, Linux, Windows

SCADA & Protocols: DNP3, DNP3-SA (Secure Authentication), Modbus RTU/TCP

Development: Git, Docker, Jupyter Notebook, pandapower, GEM5 Simulator